

LAB	PRACTICALS
25	Implementation of Sub Queries in SQL Create the following tables and perform the following queries: Part – A - Display details of students who are from computer department. <pre>SELECT * FROM STUDENT WHERE DID IN (SELECT DID FROM DEPARTMENT WHERE DNAME = 'COMPUTER');</pre> - Display name of students whose SPI is more than 8. <pre>SELECT NAME FROM STUDENT WHERE RNO IN (SELECT RNO FROM ACADEMIC WHERE SPI>8);</pre> - Display details of students of computer department who belongs to Rajkot city. <pre>SELECT * FROM STUDENT WHERE CITY = 'RAJKOT' AND DID = (SELECT DID FROM DEPARTMENT WHERE DNAME = 'COMPUTER');</pre> OR <pre>SELECT * FROM STUDENT WHERE DID IN (SELECT DID FROM DEPARTMENT WHERE DNAME = 'COMPUTER') AND CITY IN (SELECT CITY FROM STU WHERE CITY = 'RAJKOT');</pre> - Find total number of students of electrical department. <pre>SELECT COUNT(*) AS TOTAL_STUDENT FROM STUDENT WHERE DID IN (SELECT DID FROM DEPARTMENT WHERE DNAME = 'ELECTRICAL');</pre> - Display name of student who is having maximum SPI. <pre>SELECT NAME FROM STUDENT WHERE RNO IN (SELECT RNO FROM ACADEMIC WHERE SPI = (SELECT MAX(SPI) FROM ACADEMIC));</pre>

Part – B

6. Display details of students having more than 1 backlog.

```
SELECT * FROM STUDENT
WHERE RNO IN ( SELECT RNO FROM ACADEMIC
              WHERE BKLOG > 1
            );
```

7. Display name of student who is having second highest SPI.

```
SELECT NAME FROM STUDENT
WHERE RNO IN ( SELECT RNO FROM ACADEMIC
              WHERE SPI = ( SELECT MAX(SPI) FROM ACADEMIC
                           WHERE SPI < ( SELECT MAX(SPI)
                                           FROM ACADEMIC
                                         )
            )
            );
```

8. Display name of students who are either from computer department or from mechanical department.

```
SELECT NAME FROM STUDENT
WHERE DID IN ( SELECT DID FROM DEPARTMENT
              WHERE DNAME IN ('COMPUTER', 'MECHANICAL')
            );
```

Part – C

9. Display name of students who are in same department as 102 studying in.

```
SELECT NAME FROM STUDENT
WHERE DID IN ( SELECT DID FROM STU
              WHERE RNO = 102
            );
```

10. Display name of students whose SPI is more than 9 and who is from electrical department.

```
SELECT NAME FROM STUDENT
WHERE RNO IN ( SELECT RNO FROM ACADEMIC
              WHERE SPI > 9
            ) AND DID IN ( SELECT DID FROM DEPARTMENT
                          WHERE DNAME = 'ELECTRICAL'
            );
```

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Implementation of advanced JOIN operations in SQL**Create the following tables and perform the following queries:****PART – A**

--1. LIST ALL EMPLOYEES WHO EARN A SALARY GREATER THAN THE AVERAGE SALARY OF ALL EMPLOYEES.

```
SELECT * FROM EMP_DETAIL
WHERE SALARY > ( SELECT AVG(SALARY) FROM EMP_DETAIL );
```

--2. FIND THE NAMES OF EMPLOYEES WHO WORK IN THE 'SURAT' LOCATION.

```
SELECT NAME FROM EMP_DETAIL
WHERE DEPTID IN ( SELECT DEPTID FROM DEPARTMENTS
                  WHERE LOCATION = 'SURAT'
                );
```

--3. LIST ALL EMPLOYEES WHO DO NOT WORK IN THE 'RAJKOT' AND 'AHMEDABAD' LOCATION.

```
SELECT NAME FROM EMP_DETAIL
WHERE DEPTID IN ( SELECT DEPTID FROM DEPARTMENTS
                  WHERE LOCATION <> 'RAJKOT'
                  AND
                  LOCATION <> 'AHMEDABAD'
                );
```

PART – B

--4. LIST THE NAMES OF DEPARTMENTS THAT HAVE MORE THAN TWO EMPLOYEES.

```
SELECT DEPTNAME FROM DEPARTMENTS
WHERE DEPTID IN ( SELECT DEPTID FROM EMP_DETAIL
                  GROUP BY DEPTID
                  HAVING COUNT(*) > 2
                );
```

--5. LIST ALL EMPLOYEES WHO EARN THE SAME SALARY AS BOBY.

```
SELECT NAME FROM EMP_DETAIL
WHERE SALARY IN ( SELECT SALARY FROM EMP_DETAIL
                  WHERE NAME = 'BOBY'
                );
```

--6. LIST EMPLOYEES HIRED AFTER THE EMPLOYEE NAMED 'ALI'.

```
SELECT NAME FROM EMP_DETAIL
WHERE HIREDATE > ( SELECT HIREDATE FROM EMP_DETAIL
                  WHERE NAME = 'ALI'
                );
```

PART – C

--7. LIST EMPLOYEES WHOSE SALARY IS EITHER THE HIGHEST OR THE LOWEST WITHIN THE 'HR' DEPARTMENT.

```
SELECT NAME FROM EMP_DETAIL
WHERE DEPTID IN ( SELECT DEPTID FROM DEPARTMENTS
                  WHERE DEPTNAME = 'HR'
                )
AND
SALARY IN ( SELECT MIN(SALARY) AS MIN
            FROM EMP_DETAIL
            WHERE DEPTID =
              ( SELECT DEPTID FROM DEPARTMENTS
                WHERE DEPTNAME='HR'
              )
            );
```

OR

```
SELECT NAME FROM EMP_DETAIL
WHERE DEPTID IN ( SELECT DEPTID FROM DEPARTMENTS
                  WHERE DEPTNAME = 'HR'
                )
AND
SALARY IN ( SELECT MAX(SALARY) FROM EMP_DETAIL
            WHERE DEPTID IN
              ( SELECT DEPTID FROM DEPARTMENTS
                WHERE DEPTNAME = 'HR'
              )
            );
```

--8. LIST EMPLOYEES WHO WORK IN A 'SURAT' LOCATION AND HAVE A SALARY LESS THAN THE AVERAGE SALARY OF ALL EMPLOYEES.

```
SELECT NAME FROM EMP_DETAIL
WHERE DEPTID IN ( SELECT DEPTID FROM DEPARTMENTS
                  WHERE LOCATION = 'SURAT'
                )
AND
SALARY IN ( SELECT SALARY FROM EMP_DETAIL
            WHERE SALARY <
              ( SELECT AVG(SALARY) FROM EMP_DETAIL
                WHERE DEPTID IN
```

```
( SELECT DEPTID FROM DEPARTMENTS  
  WHERE LOCATION = 'SURAT'  
)  
)  
);
```

--9. LIST THE NAME AND SALARY OF ALL EMPLOYEES WHO EARN MORE THAN THE MINIMUM SALARY PAID IN THE 'IT' DEPARTMENT.

```
SELECT NAME, SALARY FROM EMP_DETAIL  
WHERE SALARY > ( SELECT MIN(SALARY) FROM EMP_DETAIL  
                  WHERE DEPTID IN (SELECT DEPTID FROM DEPARTMENTS  
                                    WHERE DEPTNAME = 'IT'  
                                )  
                );
```